MIL-B-45693(Ord)
10 February 1961
SUPERSEDING
FA(FC)-PD-126
23 April 1959

MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8639804 (OUTPUT UNIT)

1. SCOPE

1.1 This specification covers one type of output unit utilized to convert electronic signals to rotary motion for the output of the electronic analog Ballistics Computer: XM16C which forms a part of a fire control system for tanks.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment, and Tools; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

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DRAWINGS

Ordnance Corps

D8614553	Electrical Schematic
F8639804	Output Unit

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one output unit representative of production shall be made available to the Government to determine if the output unit meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The output unit shall be manufactured in accordance with Drawing F8639804, "Output Unit, " and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926.

(a) Order precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish painting and protective
(e) Electrical
(f) Workmanship

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3.5 Environmental

3.5.1 Operating temperatures.- The output unit shall meet the performance requirements of 3.9.6 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

3.5.2 Storage temperatures.- The output unit shall show no evidence of physical failure and shall meet all performance requirements of this specification at standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.5.3 Vibration.- The output unit shall be subject to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. The motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total 30 minutes). Subsequent to vibration the output unit shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.4 Humidity.- The output unit with cover in place shall be exposed to one complete humidity cycle as shown on Figure 1. Immediately following completion of the cycle, the cover shall be removed. The electronic and mechanical components shall not indicate damage or be adversely affected at which time the unit shall be checked out to meet all operational requirements of this specification (see 3.9).

3.5.5 Water resistance.- The output unit shall meet the requirements of Procedure I of the rain test of Specification MIL-F-13926.

3.6 Continuity.- The electrical circuits shall be tested for continuity and shall conform to the Schematic Drawing,D8614553.

3.7 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of output unit are subjected to an alternating current (ac) root mean square (rms) potential of 500 volts plus twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than five seconds.

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3.8 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.9 Performance.- The reference or signal voltage referred to as plus or minus ac shall signify voltages instantaneously positive or instantaneously negative when referred to a common neutral, and shall be as indicated on Schematic Drawing D8614553.

3.9.1 Accuracy.- The angular position of the output shafts shall be consistent with the values specified herein. Excitation of the follow-up potentiometer shall be 2.5 mils per volt plus or minus .05 percent nominally throughout the full output range of minus 1.0 mil to plus 55.0 mils. The scale factor shall be modified by the effective full charge to allow for variation in muzzle velocity.

3.9.1.1 Voltage slope.- When the potentiometer is energized with plus 24.25 volt ac, plus 23.75 ac and minus 1.5 volts, ac (see 6.3) through appropriate trimmer potentiometer, a slope of 2.5 mils per volt shall be achieved.

3.9.1.2 Potentiometer rotation.- When the wiper of the potentiometer is returned through a 250K 0.05 percent resistor to a servo amplifier, the servo system being energized, the angular rotation shall be related linearly to the voltage applied to a 250K 0.05 percent resistor on the input of a servo amplifier within a tolerance of ±0.1 percent.

3.9.2 Output shaft rotation.

3.9.2.1 Shaft excursion.- The output shaft for the ballistics drive shall rotate through an excursion of at least 11 rotations 72 degrees.

3.9.2.2 Counter excursion.- The superelevation counter shall cover an excursion from at least minus 1.0 mil to plus 55 mils when the requirements of 3.9.2.1 are met.

3.9.2.3 Counter indicator ballistics drive.- The counter shall indicate the angular position of the output shaft for the ballistic drive in relationship that one revolution of the output shaft is equal to 5 mils as read on the counter.

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3.9.2.4 Counter indicator superelevator.- The counter shall indicate the angular position of the output shaft for the superelevator in relationship that one revolution of the output shaft is equal to 6.25 mils as read on the counter.

3.9.3 Backlash potentiometer.- Backlash between the potentiometer and the output shaft for the ballistic drive shall not exceed 0.05 mil.

3.9.3.1 Backlash superelevator.- Backlash between the output shaft for the ballistic drive and the output shaft for the superelevator shall not exceed 0.5 mil.

3.9.4 Illumination.- With 24 volts applied to pins M and A of connector J-301 the instrument light shall supply sufficient illumination to enable the counter to be read in a darkened area.

3.9.5 Stability.- With a load of 2 inch pounds applied to the output shaft for the ballistics drive, the output unit shaft shall display stability at any point throughout the entire range of rotation.

3.9.6 Operability.- With a load of 5 inch pounds applied to the output shaft for the ballistics drive the slewing speed shall be at least 2 revolutions (10 mils) per second when checked to the requirements of 3.5.1.

3.9.7 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

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4.4 Production model.- Unless otherwise specified in the contract, one output unit shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production model failure.- Should the production model fail to pass the prescribed examinations and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

4.5.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.2 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the output unit. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.5.2.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

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TABLE I- SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.2.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.7 Acceptance tests.

4.7.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%		Req't. Par.	QA. Par.	Test Method
101.	Operating temperature	3.5.1	4.9.1.4	Device, Special Testing (DST)
102.	Physical failure (post vibration)	3.5.3	4.9.1.2	Visual

MINOR: NONE DEFINED

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Continuity	3.6	4.9.1.7	Standard Measuring Equipment (SME)
102.	Dielectric strength	3.7	4.9.1.8	SME
103.	Insulation resistance	3.8	4.9.1.9	SME
104.	Accuracy	3.9.1	4.9.2.1	SME
105.	Voltage slope	3.9.1.1	4.9.2.2	SME
106.	Potentiometer wiper rotation	3.9.1.2	4.9.2.3	SME
107.	Shaft excursion	3.9.2.1	4.9.2.4	SME
108.	Counter, excursion	3.9.2.2	4.9.2.5	Visual
109.	Counter, indicator (ballistics drive)	3.9.2.3	4.9.2.6	Tactile-Visual
110.	Counter, indicator (superelevator)	3.9.2.4	4.9.2.7	Tactile-Visual
111.	Backlash potentiometer	3.9.3	4.9.2.8	Device, Special Testing (DST)
112.	Backlash superelevator	3.9.3.1	4.9.2.9	DST
113.	Illumination	3.9.4	4.9.3	Visual
114.	Stability	3.9.5	4.9.4	DST
115.	Operability	3.9.6	4.9.5	DST
116.	Identification marking	3.9.7	4.9.6	Visual

MINOR: NONE DEFINED

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4.8 Inspection equipment.

4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.8.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements shall be capable of measuring these values within an accuracy of 10 percent of the specified tolerance, except in those cases specified otherwise. In cases where the 10 percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.2 Accuracy requirements for special testing devices.

4.8.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion at a uniformly variable frequency between the limits of 10 and 55 cycles per second and return to 10 cycles. It shall have an amplitude of 0.03 inch (0.06 inch maximum excursion) and shall be equipped with a medium capable of holding the unit securely to the vibrating machine and applying the produced motion in each of 3 mutually perpendicular directions. The accuracy of the device shall be in accordance with tolerances governing vibration test condition measurements contained in Specification MIL-F-13926.

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4.8.2.2 Device to test potentiometer backlash and superelevation backlash.- A special testing device capable of measuring the backlash between the potentiometer and the output shaft for the ballistic drive shall be accurate within plus or minus .05 mil. Provisions shall be made to measure the backlash between the output shaft for the ballistic drive and the output shaft for the superelevator, the accuracy of which shall be plus or minus .05 mil.

4.8.2.3 Device to test stability and operability.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to ten inch-pounds, the accuracy of which shall be within plus or minus one-quarter inch-pound throughout the entire zone of excursion. The device shall be equipped with an adapter to accommodate rotation of the output shaft for the ballistics drive in both clockwise and counterclockwise directions.

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Vibration.- The output unit shall be vibrated in accordance with Standard MIL-STD-202, Method 201A, except the motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total 30 minutes).

4.9.1.2 Physical failure, post vibration.- Subsequent to vibration, the output unit shall be subjected to a visual and tactile examination for damage. The unit then shall be subjected to all the tests specified.

4.9.1.3 Storage temperature.- This test shall be conducted in accordance with Procedure II of MIL-F-13926 to the conditions specified. Upon completion of the specified exposure, the unit shall be inspected for damage, and then subjected to the test specified in 4.9.2, and shall meet the requirements specified. This test shall be performed on the production model only.

4.9.1.4 Operating temperature.- When thermal equilibrium of the output unit is reached at plus 150 degrees Fahrenheit and minus 40 degrees Fahrenheit, the unit shall be subjected to the tests in 4.9.2 and shall meet the requirements specified.

4.9.1.5 Water resistance (rain).- This test shall be conducted in accordance with Procedure I of MIL-F-13926. The output unit shall show no evidence of water penetration as a result of this test. This test shall be performed on the production model only.

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4.9.1.6 Humidity.- The output unit to be tested shall be placed in the test chamber in its normal operating position and exposed to one complete humidity condensation cycle as specified. Distilled or demineralized water having a pH value between 6.5 and 7.5 at 77 degrees Fahrenheit shall be used to obtain the desired humidity. The unit under test shall be placed no closer than 6 inches from the sides of the chamber, and shall not be subject to radiant heat. Upon completion of the humidity exposure, the unit shall show no evidence of damage. It shall then be subjected to the tests in 4.9.2, and shall meet the requirements specified. This test shall be performed on the production model only.

4.9.1.7 Continuity.- A continuity check of the output unit shall be made using standard measuring equipment to determine conformance to the schematic drawing.

4.9.1.8 Dielectric strength.- The output unit shall be tested in accordance with Method 301 of MIL-STD-202, and shall meet the requirements specified.

4.9.1.9 Insulation resistance.- Immediately after satisfactorily withstanding the tests for dielectric strength, the output unit shall be given the insulation resistance test in accordance with Method 302, test condition B of MIL-STD-202, and shall meet the requirements specified.

4.9.2 Performance.

4.9.2.1 Accuracy.- The angular position of the output shafts shall be checked for consistency with the excitation of the follow-up potentiometer and shall comply with the requirements and allowable tolerance.

4.9.2.2 Voltage slope.- With the potentiometer energized as specified, the voltage slope shall be measured with standard measuring equipment and shall comply with the requirements.

4.9.2.3 Potentiometer wiper rotation.- The linear relationship of the angular rotation of the potentiometer wiper shall be ascertained by means of standard measuring equipment and shall be within the tolerance specified.

4.9.2.4 Shaft excursion.- The minimum amount of rotation of the output shaft for the ballistics drive shall be measured by means of standard measuring equipment and must be not greater than the amount specified.

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4.9.2.5 Counter excursion.- During the preceeding test, a visual observation shall be made to be assured that the excursions requirements have been met.

4.9.2.6 Counter indicator, ballistics drive.- The ratio of mils to revolutions between the counter indicator and the ballistics drive output shaft shall be checked for conformance with the requirements.

4.9.2.7 Counter indicator, superelevator.- The ratio of mils to revolutions between the output shaft and the counter indicator shall be checked for conformance with the requirements.

4.9.2.8 Backlash, potentiometer.- The measured backlash between the potentiometer and the output shaft shall not exceed the allowable tolerance.

4.9.2.9 Backlash, superelevator.- The measured backlash between the output shaft for the ballistics drive and the output shaft for the superelevator shall not exceed the allowable tolerance.

4.9.3 Illumination.- The instrument light shall be energized as directed and the counter visually checked in a darkened area. The counter must be legible under these conditions.

4.9.4 Stability.- Stability shall be checked under the load and rotation requirements specified.

4.9.5 Operability.- Operability shall be checked under the load and temperature requirements.

4.9.6 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The output unit shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-F-14619/P11746. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- This output unit is primarly intended as a functional component of the electronic analog Ballistics Computer: XM16C. The unit is designed to accept servo controlled signals from a computing unit representative of superelevation with corrections added for droop, jump and cant and translate the electronic signals to rotary motion for introduction to a ballistic drive and a superelevator.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provision for production model including date of delivery (see 3.1)

6.3 AC voltage polarity.- Reference to plus ac voltage and minus ac voltage are instantaneous voltages in relation to each other.

6.4 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.5 Definitions

6.5.1 Device, special testing (DST).- Special testing devices are designed to check function of an assembled unit.

6.5.2 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.5.3 Tactile.- Skillful manipulation or sensing.

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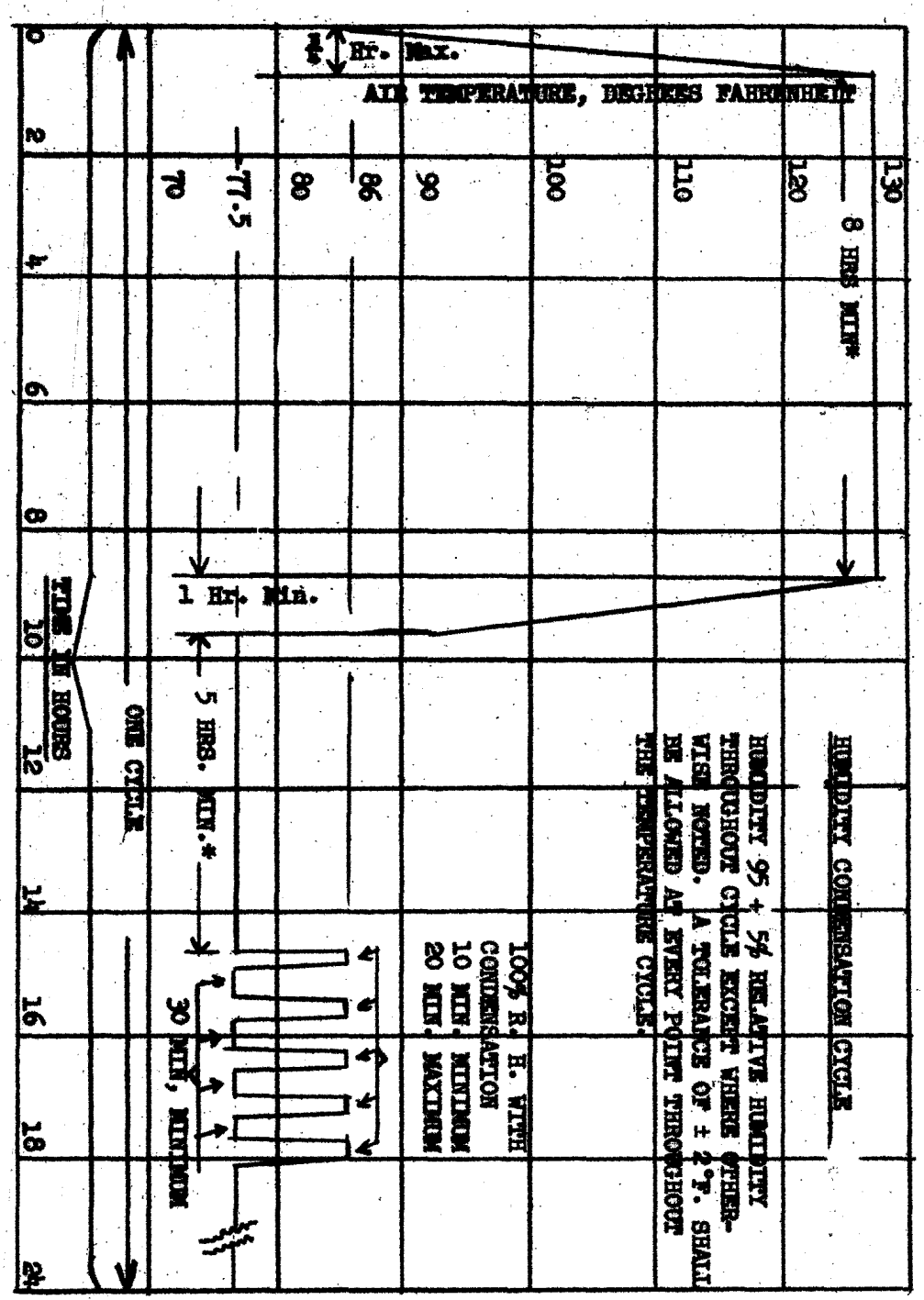


FIGURE 1

* OR UNTIL EQUILIBRIUM STABILIZATION (WHICHEVER IS LONGER)

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